Generative AI Guidelines

The following guidelines were built collectively as a class. Together we discussed various topics that covered the PROS and CONS of using gen-AI, potential ethical issues of the tool, and any effective strategies to use the tool.

Note: This is not a policy document on gen-AI. The aim of this document is to provide recommendations and advice that we all have collectively agreed upon and have discussed about on how we can use gen-AI while considering its potential, limitations, and ethical implications. This document is subject to change—how we use and think about gen-AI could change as the tool itself is rapidly becoming more advanced.

**** No generative AI technologies were used while creating this document. ****

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I. Pros & Cons of Using Gen-AI: Classroom Discussion

In class, we discussed what are the PROS and Cons of using gen-AI (in general). Below are screenshots of a couple responses generated by the class. A general theme for the PROS is that it helps you with brainstorming—coming up with ideas on how to start a problem could be challenging however with gen-AI, it can help you jump start. A general theme for the CONS is that despite helping with offloading mental effort, it can potentially decrease your ability to critically think and solve problems.

Cons: Whenever I seldom use it, it tends to be wrong--especially if it involves something like algebra.

Gen-AI can be very useful as a resource when attempting to solve problems. If used correctly, Gen AI can provide a means to learning new concepts and practicing skills.

Cons: Less practice with creative problem solving.

Unreliable at times. Need to double check what it gives you

Pro: Ideas you didn't think of Cons: Accidental plagiarism

Pros: Good Fun Makes me happy Expedites coding generates funny images Cons: People make fun of you makes it harder to learn I am not learning coding as fast

Pro: Allows you to ask questions and learn in a more conversational way if you don't have someone else to ask

con: it can decrease our ability to learn and think critically

Pro: Does your work for you Con: Does your work for you

Offloading mental effort

Debugging is faster

II. Ethical Issues: Classroom Discussion

In class, we also identified the potential ethical issues revolving around gen-AI. Here are some big ethical issues that were listed by the class:

1. Academic integrity

Take the Liberal Arts college art competition winner, for example. They used AI and they won first place. Additionally, students use it to write their essays for them and cheat on tests

Makes cheating easy 😈

People are submitting work for class as their own, without having done the work and using a program that uses the work of others without compensation.

2. Transparency

Data used for training generally used without authors' consent, potentially breaches copyright/IP

Often trained on copyrighted data. It has monetary value and is being used for the same purpose as the initial work, so it is copyright theft if done without permission.

Where the training data sets come from

3. Trustworthiness

Creating false information that is hard to distinguish from the truth

Theft or spreading misinformation

4. Privacy

It can remember sensitive information

Generative AI raises issues like misinformation, bias, job loss, IP concerns, privacy risks, and unclear accountability in how its outputs are used.

5. Algorithmic and data biases

Kinda racist

bias

People be spreading dis/misinformation

6. Environmental Impact

computing power impacts on the environment

III. Level of Usage

What are reasonable and unreasonable uses of generative AI in the classroom?

Gen-AI is a tool that can be utilized for many things. In our classroom discussion, several worries were raised especially on the state of our critical thinking skills and learning ability. **Thus, the following table identifies the reasonable and unreasonable usages that we came up with to help us address our worries without completely rejecting gen-AI—each column incorporates what gen-AI can/cannot do but also what we can do as the user to mediate our own learning!**

Reasonable	Unreasonable
It's reasonable to use gen-AI to... <u>Debug</u> • Identify the errors in your code • Using gen-AI to explain what the errors mean <u>Clarify</u> • Help clarify what the instructions are asking you to do • Simplify or explain a concept or function <u>Generate Code</u> • Generate code that you had trouble writing on your own, as long as you explain what the code is doing to demonstrate your understanding of what was generated	<u>It's unreasonable to use gen-AI to...</u> <u>Copy & Pasting</u> • Directly copy & paste the instructions from the assignment into gen-AI without incorporating your own question(s) into the prompt • Directly copy & paste the generated answers by gen-AI without adding any explanation of your understanding into the solution

IV. Documenting Gen-AI Usage

How should we document the usage of gen-AI in our classwork? As a class, we recognized the importance of transparency with our uses of gen-AI. Therefore, the following table presents suggestions on how we can be transparent and communicate when, where, and what it means to credit gen-AI in our assignments.

Commenting	
When should we declare usage gen-AI?	When the generated output by gen-AI contributes to your code. Contribution to the code may include the following: • Correcting your code • Generating code that you had difficulty with in writing yourself • Generating code for brainstorming purposes (i.e., getting inspiration from the gen-AI on how to start writing the code)
Where should we declare the usage of gen-AI?	Comment at the beginning of the section if: You have used it for a large section of the code and/or consistently for a particular reason (e.g., consistently using ChatGPT to debug your code throughout the section) Comment next to the line of code if: An individual line was partially (even one word of syntax) or entirely generated by gen-AI.
What do we comment on?	A short description of which gen-AI software you used & how you used it to help you • *Ex: I used ChatGPT to help debug my error code that was about XYZ...I applied the generated output by...* Your understanding of the code & what gen-AI produced • *Ex: This code is about XYZ...Gemini generated a code that does XYZ and from my understanding this means...etc.*

V. Strategies

What are the strategies we can use to prompt gen-AI and use the AI-generated content in meaningful ways for our learning? These strategies were created and compiled by the class. Keep in mind, these strategies could potentially change as gen-AI is rapidly advancing.

Strategy #1: Ask specific questions when you're prompting.

This may include being self-aware of what you need help with. Providing detailed prompts to gen-AI may help you resolve your issue faster. Additionally, more detailed and specific prompts may improve the conversation between you and the gen-AI as it has a better idea of what you need help with.

Note: Using keywords *may* be helpful but it does require you to know what keywords to use. Additionally, chatbots are not like google search engines where keywords do come in handy, as chatbots will need more specific context to cater to your needs.

Strategy #2: Specifically ask generative AI to explain in simple/layman's terms.

This is helpful if the content produced by gen-AI is something you have a difficult time understanding. Having the gen-AI to rephrase it in terms that you can understand may help! Additionally, consider asking it to explain with different representations (diagrams, analogies, etc.).

Strategy #3: Ask for explanations and/or examples rather than prompting it to produce the exact code you need.

This encourages you to learn how to construct the code on your own with some guidance by gen-AI! The examples can be used as a way to brainstorm in constructing the code whereas the explanations are useful in providing a clearer definition of what the concepts aim to do!

Strategy #4: Be aware, you can tell the gen-AI that it is wrong, but it still might not correct itself properly.

To check this, ask a peer, LA, or the professor to help interpret the output from the gen-AI. You can also check other sources (e.g., websites) to help you check what's been generated.